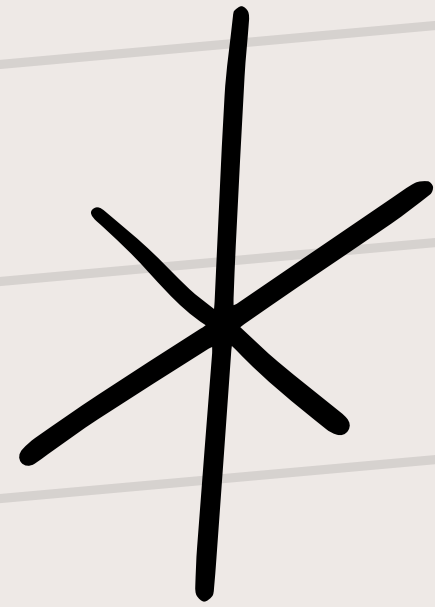
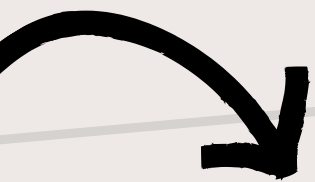




Project Update

Group 9



Regional English Accent Robustness in Speech Translation

Direct and Cascaded English-to-Chinese systems

Problem

Speech translation quality may vary across speakers with different self-reported regional English accents

Why it matters

This could cause uneven performances for some speaker groups, reducing reliability and accessibility in translation tools

What Prior work shows

Accent affects ASR

Accent-related ASR disparities span regional, dialectal and second-language English, including in both Whisper and SeamlessM4T evaluations (DiChristofano et al., 2022; Jahan et al., 2025; Swain et al., 2026).

ASR errors can propagate

Cascaded translation is sensitive to transcript errors (Sperber et al., 2019). 6.79 to 10.64 BLEU lost is reported when ASR output replaces clean transcripts for an Vietnamese to English study. (Nhuyen et al., 2026)

Ranking depends on setup

Direct vs. Cascaded preference score changes with language pair, data, references and metric, there is no single settled winner between the two (Bentivogli et al., 2021; Anastasopoulos et al., 2021; Etchegoyhen et al., 2022).

The gap

No study in our 27-paper review combines, in one experiment: controlled regional-English accent groups, the same utterances and references, both Direct and Cascaded systems, and final translation quality as the main outcome.

Progress update

Research Question

How does regional English accent affect the translation quality of Direct and Cascaded speech-to-text translation systems across seven self-reported accent groups?

Objective and status

1

Construct a balanced evaluation set

Seven self-reported accent groups, each 600 clips = 4,200 clips

COMPLETE

2

Compare two ST systems on the same speech

Direct: Seamless M4T v2 vs. Cascaded: Whisper +NLLB

COMPLETE

3

Measure translation quality across groups

Corpus BLEU, chrF and chrF++ vs. Chinese references, per group

INITIAL RESULTS

4

Measure accent related ASR variation

Normalised WER per group

INITIAL RESULTS

5

Link ASR errors to Cascaded degradation

Association analysis of WER vs. Cascade translation quality. (Nota causal claim)

PENDING

Expected Contribution

A paired English-to-Chinese comparison on the same 4,200 clip-reference pairs, organised into seven equal-sized groups by self-reported accent label.

Boundary

A descriptive comparison. It does not by itself prove that accent or architecture caused a score difference.

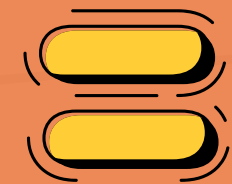
What Changed Since Our Proposal?

**Key Change**

Proposal was
5 Groups
3000 clips in total

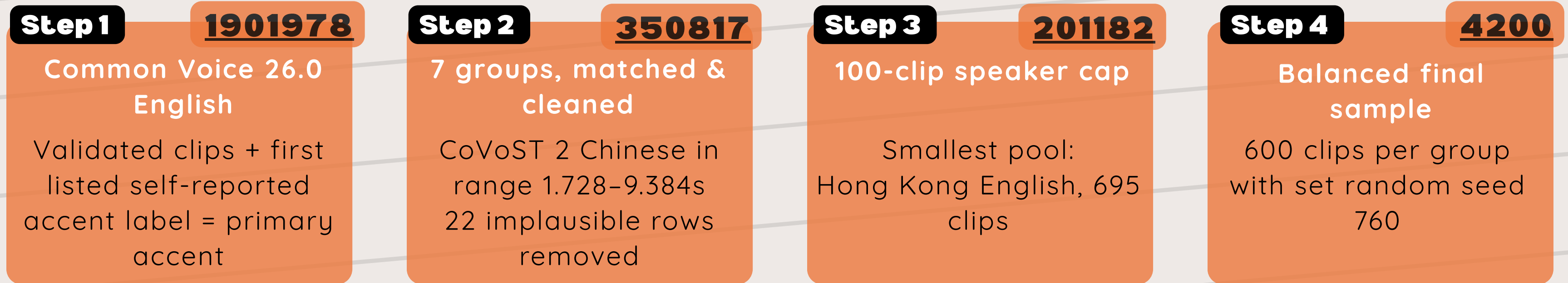


England English & United States English
Added as native reference groups. A six-candidate audit selected them on both pool size and speaker diversity

**Now**
7 Groups
4200 clips in total

<u>AREA</u>	<u>ORIGINAL PLAN</u>	<u>CURRENT IMPLEMENTATION</u>	<u>WHY</u>
 Dataset build	Filter steps + Colab verification notebook (5 groups)	Preprocessing & audio-extraction notebooks in the Git repo	Full build, incl. audio extraction, now versioned and reproducible
 Cascade execution	Conceptual Whisper + NLLB cascade	Separate Whisper and NLLB passes	Lower GPU-memory pressure; cascaded pipeline preserved
 Model versions	Model names, no frozen revisions	Exact model revisions recorded	Completed runs are reproducible
 Evaluation	Evaluation planned after inference	WER, BLEU, chrF and chrF++ computed on all 4,200 clips	Early quantitative evidence for this update

Dataset and Evaluation Set Construction



Why N = 600?

600 sampled

695

Hong Kong English had the smallest pool after the speaker cap (695 clips). 600 is the largest round N every group could supply with a margin, and the 35-clip timing pilot confirmed 7×600 was feasible.

Why England & United States English?

Best of six native candidates on both criteria:

42k -127k

clips after speaker cap

$\leq 4.6\%$

clips from any single speaker

Final Evaluation Set

4200 clips = 7 accent group * 600

- **Native reference:** England, United States
- **Regional:** Filipino, Hong Kong, India & South Asia, Malaysian, Southern African
- **Comparability limit:** groups share no identical sentences or speakers, and accent labels are self-reported.

1586

Speakers

5.39h

Of speech

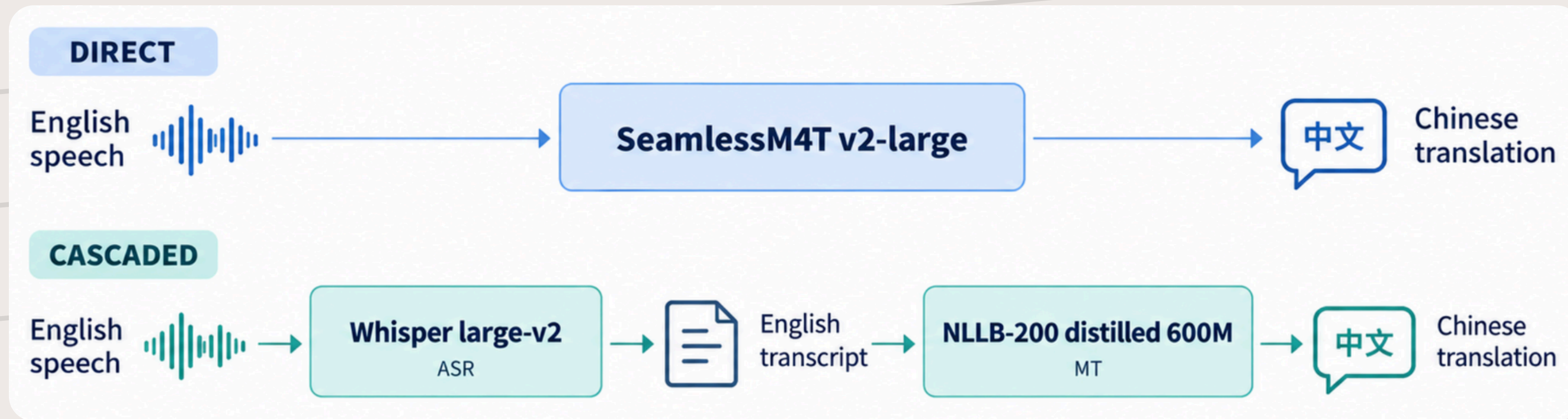
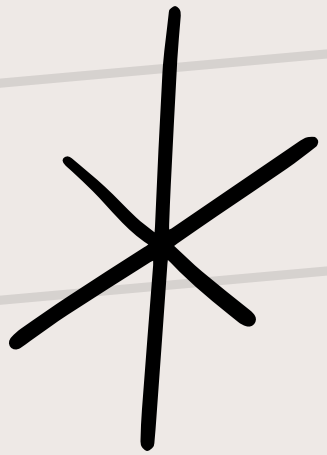
4.62s

Mean clip

0

missing refs

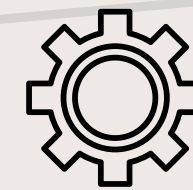
Experimental Design & Computational Setup



Shared conditions

Same 4,200 clip-reference pairs

- 16 kHz mono audio | Greedy decoding | 256-token limit
- No fine-tuning, accent adaptation or manual correction



Google Colab, NVIDIA T4 (14.56 GB)
CUDA 12.8 and FP16 inference
PyTorch 2.11.0 and Transformers 4.57.6

	Direct	Cascaded
Total runtime	3,898.6 s (~65.0 min)	3,903.4 s (~65.1 min)
Mean / clip	0.928 s	0.929 s
Peak GPU memory	2.876 GB	3.159 GB



System Level Comparison

Model families and capacities differ

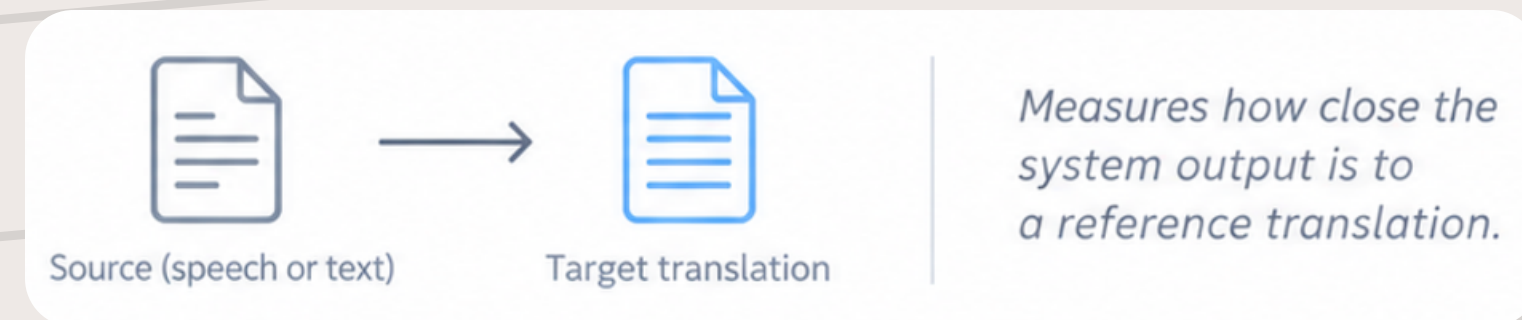
Evaluation Protocol

Translation Quality

Q: How good is the Chinese translation?

BLEU, chrF and chrF++

Higher is better



Q: Does performance vary by group?

Overall and per-group corpus scores

Next Step

Q: Are differences reliable?

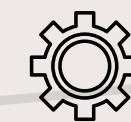
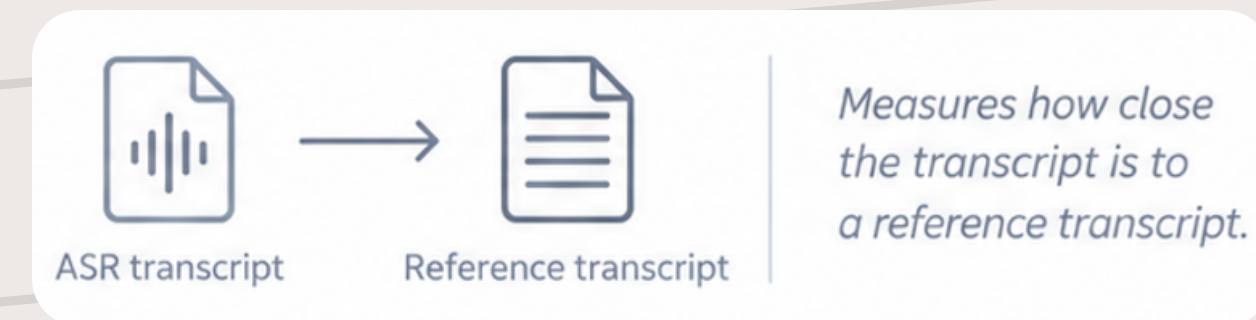
Paired bootstrap uncertainty analysis.

ASR Accuracy

Q: How accurate is Cascaded ASR?

Normalised WER

Lower is better



Frozen pretrained models
One fixed evaluation set



No training or fine-tuning
No train/validation/test split

Two Visible Error Patterns

Qualitative Output Audit

1

Direct: Unknown Token

sample_0002: "Hall-Jones"

霍尔-<unk>斯

Cascade: 霍尔-乔恩斯

2

Cascade: ASR error Propagation

sample_0001: "side show in the circus"

我们谈论了马戏团的侧鞋

Whisper heard "side shoe"

Direct: 我们谈到了马戏团的副表演 (Side Performance)



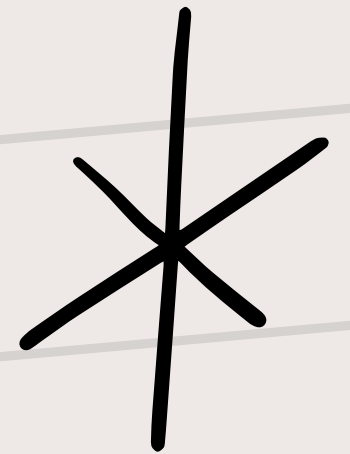
12.1%

508/4,200 Direct outputs contain <unk>

Subset BLEU 25.64 with <unk> | BLEU 40.67 without <unk>



Subsets differ in content and difficulty.
BLEU gap is descriptive, not a causal token penalty.



Quantitative Evaluation

Preliminary Results — Direct vs Cascaded

Overall Corpus Result — 4,200 clips

Direct BLEU: 38.86

Cascaded BLEU: 29.11

+9.75 Direct advantage

Normalised corpus WER (Cascade only): 5.97%

Direct Advantage by Metric

Metric	Direct Advantage (gap range)
BLEU	+7.69 to +11.46
chrF	+6.25 to +10.09
chrF++	+5.44 to +7.97

Direct leads on all 21 accent × metric comparisons — best case Filipino, narrowest lead India & South Asia.

21/21

Direct wins every accent × metric comparison

7 accent groups × 3 translation metrics (BLEU, chrF, chrF++) — Direct ahead in all 21.

Cascaded WER: 4.3% (US English) to 7.6% (Malaysian)

Ranking doesn't match BLEU / chrF / chrF++ ranking — accuracy alone doesn't explain translation quality.

Single corpus-level point estimates — no confidence intervals or significance tests yet. Differences are descriptive; paired bootstrap uncertainty analysis is still pending.

Existing Literature vs Our Results

How Our Preliminary Evidence Compares to the Review

Published Finding	Our Evidence	Interpretation
Accent-related ASR disparities (DiChristofano et al. (2022) [12]; Jahan et al. (2025) [13]; Swain et al. (2026) [14])	Cascaded WER: 4.3% (US English) to 7.6% (Malaysian)	Consistent with prior work ; but untested; the spread alone doesn't confirm an accent effect.
ASR errors can propagate (Sperber et al. (2019) [15]; Nguyen et al. (2026) [16]; Havare et al. (2026) [21])	Malaysian & India/South Asia: highest WER + lowest Cascaded BLEU. US English breaks the pattern.	Compatible with propagation, but WER alone doesn't explain translation quality ; formal test pending.
Closest accent-aware studies use different designs (Seamless Communication et al. (2023) [3]; Robinson et al. (2024) [6]; Papi et al. (2026) [17])	Direct scores higher in all 21 accent × metric comparisons	Our paired design fills part of this gap ; pretraining overlap and unequal capacity block an architecture claim.
Rankings depend on setup (Bentivogli et al. (2021) [18]; Anastasopoulos et al. (2021) [19]; Etchegoyhen et al. (2022) [20])	Our accent-level BLEU gaps: +7.69 to +11.46 Direct	Not directly comparable ; the large gap motivates confound checks, not a claim of inherent superiority.

Contribution: a paired Direct-vs-Cascaded English → Chinese evaluation on the same 4,200 clip-reference pairs, across seven balanced accent groups.

WER diagnoses only the Cascaded ASR stage — cross-system conclusions rely on final translation metrics. Paired, speaker-aware uncertainty is still required before claiming a confirmed accent effect.

Progress, Challenges and Adaptations

Timeline

&

Status

Week 1-2

Prepare Data and Run Pipelines

Complete: Expanded to 4,200 clips and finished both systems

Week 3

Compute WER and Translation Metrics

Complete: initial WER, BLEU, chrF and chrF++

Week 4

Statistical and Error Analysis

In progress: descriptive audit complete. Paired uncertainty and association analysis remain

Week 5

Interpret and Deliver

In progress: progress presentation now. Final figures and report come next

Key Adaptations:

- Staged Whisper and NLLB plus checkpointing to fit the T4 and protect long runs.
- Streamed only selected files from the 88 GB archive instead of unpacking the full dataset.
- Removed Common Voice audio and raw identifiers from the public repository, then purged Git history. Audio remains private and shared metadata is pseudonymised.



Conclusions and Next Steps

Conclusion

- Direct is higher in all 21 preliminary group-metric comparisons, despite its visible <unk> problem.
- Cascaded WER varies by group, but WER ranking does not fully explain translation ranking.

The evidence supports a **preliminary system-level difference**, not a confirmed causal accent or architecture effect.

Next Steps

1

Add paired bootstrap uncertainty and formal WER-translation association analysis

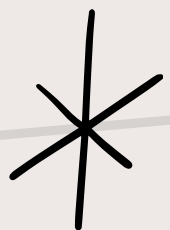
2

Finalise reproducibility details, sensitivity checks, figures and the written report

* Team Contributions

Nathan Naylin	Pilot/timing, Direct and Cascaded pipeline runs, and dataset literature review
Zhitong Li	Accent-ASR literature review and problem motivation
Arizona Xing	Data preprocesing and data extraction, pilot and full extraction, and architecture literature review
Patricia Jennesha	Accent-ST and error-propagation review, research-gap synthesis and interpretation
Jidni Mayukh	Initial evaluation results (WER, BLEU, chrF/chrF++), and evaluation metrics literature review

Our Project Github Repo Link



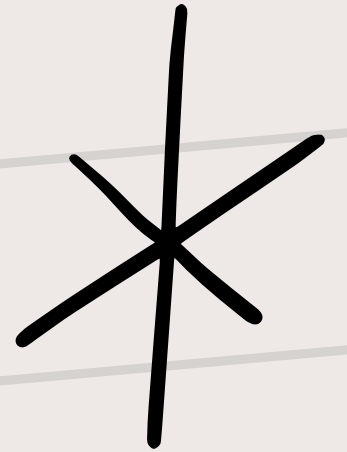
Implementation: Python and Google Colab notebooks cover preprocessing, targeted audio extraction, Direct and Cascaded inference, checkpointing, and output validation.

<https://github.com/Nlhmmh/cs760-accent-st-robustness-ml-research>



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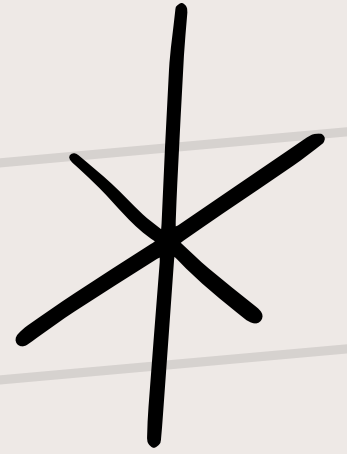
Datasets, models and evaluation methods



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Literature compared with our findings



[12] DiChristofano, A., Shuster, H., Chandra, S., & Patwari, N. (2022). Global performance disparities between English-language accents in automatic speech recognition. arXiv:2208.01157. <https://doi.org/10.48550/arXiv.2208.01157>

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GenAI Use

ChatGPT/Codex assisted with auditing, structure and wording.

The team is responsible for the design, implementation, verification and conclusions.



**Thank
You**

